Individual Case #1

BAN 120: Analytics Case Studies

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Case Notes

Location: Waterloo Ontario

Problem: Invasive plant affecting crops within the region

Plant Survival Requirements:

- Won't survive more than **4 days of temperatures less than -15°C** per year
- Requires substantial heat to survive through the growing season (at least **2500 growing degree days per year**)

3 Climate Projection scenarios:

1. Low Emissions RCP 2.6
2. Moderate Emissions RCP 4.5
3. High Emissions RCP 8.5

The local agricultural group has asked you to provide a recommendation/insight about what to do with their crops under three different climate projection scenarios.

What is Representative Concentration Pathways (RCPs)?

In simple terms, think of RCP as a story about how much pollution there will be in the future. It tells us how much greenhouse gases (like carbon dioxide) will be in the air based on different scenarios. It's like predicting how dirty the air will be in different possible futures.

3 Climate Projection scenarios in our case:

Low Emissions RCP 2.6

Moderate Emissions RCP 4.5

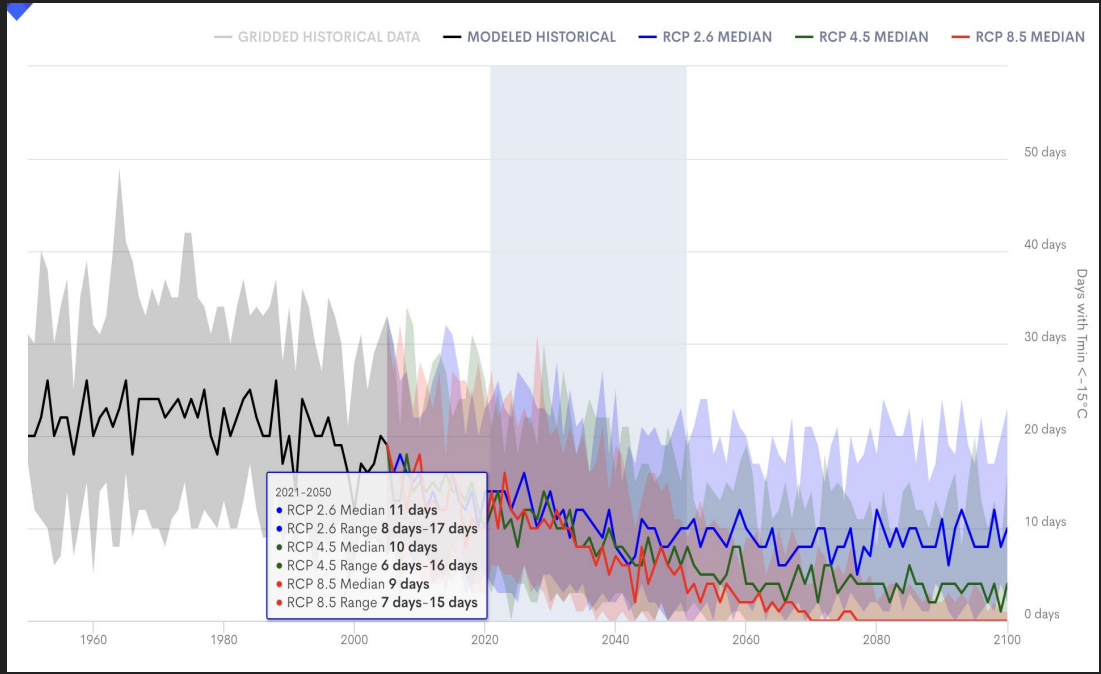
High Emissions RCP 8.5

Days with minimum temperature (Tmin) < -15°c

Looking at the years of 2021-2050

We can observe that:

- In RCP 2.6, the average number of days having temperature less than -15°c is **11 days**
- In RCP 4.5, the average number of days having temperature less than -15°c is **10 days**
- In RCP 8.5, the average number of days having temperature less than -15°c is **9 days**



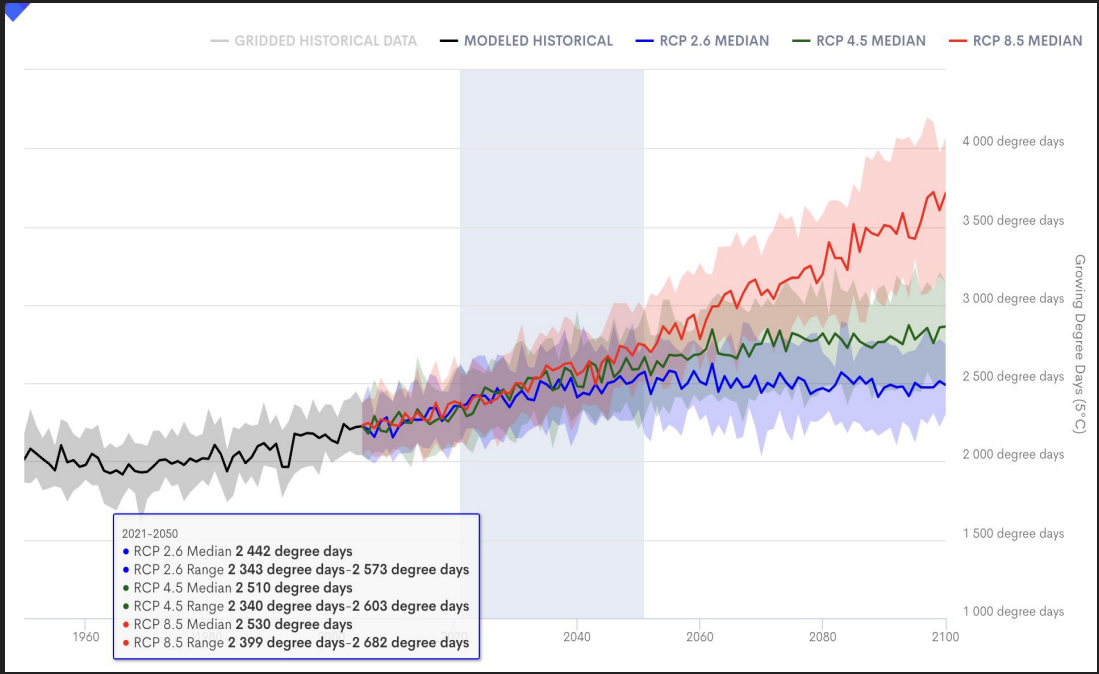
In all 3 cases, the plant won't be able to survive because it won't survive if more than 4 days of temperatures are less than -15°C per year

Growing Degree Days (5°c)

Looking at the years of 2021-2050

We can observe that:

- In RCP 2.6, the average number of Growing Degree Days is **2442 days**
- In RCP 4.5, the average number of Growing Degree Days is **2510 days**
- In RCP 8.5, the average number of Growing Degree Days is **2530 days**



The plant won't be able to survive in case 1, because it requires at least 2500 growing degree days per year. However, in case 2 and 3, it will be able to survive.

Recommendations - RCP 2.6 scenario

Where the average number of days with temperatures below -15°C is 11 days and the average number of Growing Degree Days is 2442 days:

1. Consider selecting crop varieties or species that are more tolerant to colder temperatures.
2. Promote crop rotation and diversification to minimize the spread of the invasive plant and reduce its impact on crops.
3. Implement integrated pest management strategies to detect and control the invasive plant effectively.

Recommendations - RCP 4.5 scenario

Where the average number of days with temperatures below -15°C is 10 days and the average number of Growing Degree Days is 2510 days:

1. Continue implementing the same measures as in RCP 2.6, as the conditions are similar.
2. Continue utilizing crop varieties that are resilient to colder temperatures and less susceptible to the invasive plant.
3. Focus on optimizing crop management practices to maximize productivity during the growing season with the available Growing Degree Days.
4. Implement early detection and rapid response strategies to identify and eliminate the invasive plant before it spreads extensively.
5. Consider implementing sustainable farming practices that enhance soil health and promote crop resilience.

Recommendations - RCP 8.5 scenario

Where the average number of days with temperatures below -15°C is 9 days and the average number of Growing Degree Days is 2530 days:

1. Assess the potential impact of a reduced number of extremely cold days on the survivability of the invasive plant and adjust control measures accordingly.
2. Implement strategies to manage the potential spread of the invasive plant by monitoring and removing any infested plants in its early stages.
3. Continuously assess and adapt crop choices and management practices to ensure resilience in the face of changing climate conditions.
4. Maximize the utilization of the available 2530 growing degree days to optimize crop growth and productivity.
5. Explore additional mitigation strategies such as targeted herbicide application or biological control methods to manage the invasive plant effectively.

Thank you